THE LOWER SHRIEK ALONG A FIBRED CATEGORY IS COMPUTED BY THE COLIMIT OF A SHEAF OVER THE FIBRE CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the construction of the restriction maps from pullback functors and strongly cartesian morphisms, the reduction to the adjunction identity for morphism sets, and the identification of both sides through pi^{-1}G(U) = G(p(U)).

Editorial note. The label lemma-compute-pi-shriek names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement refers to the standing situation of a fibred category; its three cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Situation of a fibred category over a site and the adjoint pair (pi_!, pi^{-1}). Same chapter, the standing situation of a fibred category p : C -> D and the lower shriek pi_! left adjoint to pi^{-1}. Supplies the functor whose value on an abelian sheaf is computed.
- Pullback functors between fibre categories and strongly cartesian morphisms. Categories, the definition of the pullback functor h^* : C_V -> C_{V'} along a morphism h : V' -> V and the strongly cartesian morphism h^*U -> U covering h. Supplies the transformations of functors whose colimits give the restriction maps of the presheaf in the statement.
- Functoriality of colimits. Categories, the lemma producing a map between colimits from a transformation of the diagrams. Supplies the existence of the restriction maps H(V) -> H(V') along h.

Source reference. Original definition 02XN: sources/stacks-chow/tags.txt, source lines 3846–3846. Prerequisite pointer invoked at proof line(s) 9977; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 002K: sources/stacks-chow/tags.txt, source lines 101–101. Prerequisite pointer invoked at proof line(s) 9988; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08P8: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9701–9701. Prerequisite pointer invoked at proof line(s) 9962; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Assumptions and notation as in Situation 08P8. For $\mathcal{F}$ in $Ab(\mathcal{C})$ the sheaf $\pi_!\mathcal{F}$ is the sheaf associated to the presheaf*

$$V \longmapsto \operatorname{colim}_{\mathcal{C}_V^{opp}} \mathcal{F}|_{\mathcal{C}_V}$$

with restriction maps as indicated in the proof.

Proof. Denote $\mathcal{H}$ be the rule of the lemma. For a morphism $h : V' \to V$ of $\mathcal{D}$ there is a pullback functor $h^* : \mathcal{C}_V \to \mathcal{C}_{V'}$ of fibre categories (Categories, Definition 02XN). Moreover for $U \in \mathrm{Ob}(\mathcal{C}_V)$ there is a strongly cartesian morphism $h^*U \to U$ covering h. Restriction along these strongly cartesian morphisms defines a transformation of functors

$$\mathcal{F}|_{\mathcal{C}_V} \longrightarrow \mathcal{F}|_{\mathcal{C}_{V'}} \circ h^*.$$

Hence a map $\mathcal{H}(V) \to \mathcal{H}(V')$ between colimits, see Categories, Lemma 002K.

To prove the lemma we show that

$$\mathrm{Mor}_{PSh(\mathcal{D})}(\mathcal{H}, \mathcal{G}) = \mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{F}, \pi^{-1}\mathcal{G})$$

for every sheaf $\mathcal{G}$ on $\mathcal{C}$. An element of the left hand side is a compatible system of maps $\mathcal{F}(U) \to \mathcal{G}(p(U))$ for all U in $\mathcal{C}$. Since $\pi^{-1}\mathcal{G}(U) = \mathcal{G}(p(U))$ by our choice of topology on $\mathcal{C}$ we see the same thing is true for the right hand side and we win. $\square$